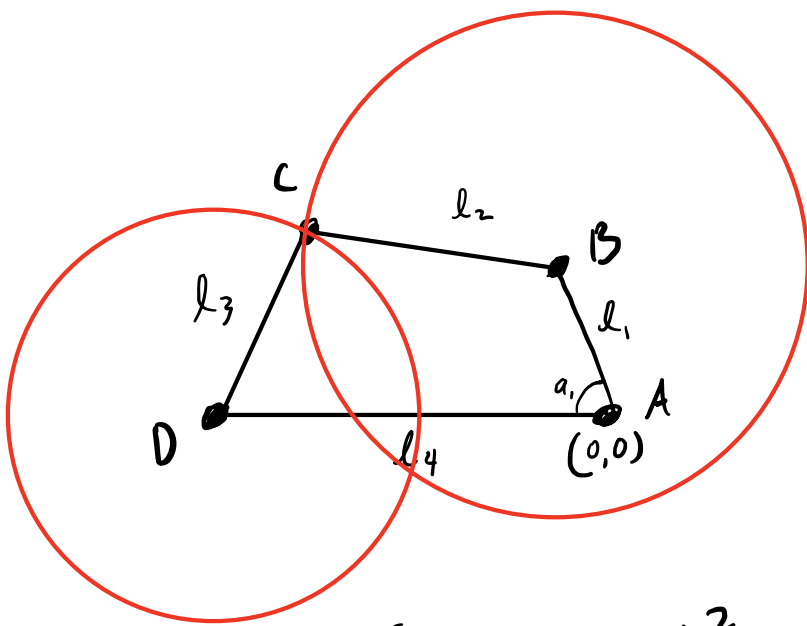


Solving For Vertex Positions Based on Servo Position



$$B: \begin{pmatrix} -\cos(a_1)l_1 & \sin(a_1)l_1 \end{pmatrix}$$

$$C: (x+l_4)^2 + y^2 = l_3^2$$

$$(x + \cos(a_1)l_1)^2 + (y - \sin(a_1)l_1)^2 = l_2^2$$

$$\overset{e}{(x+l_4)^2} + y^2 = l_3^2$$

$$\underbrace{(x + \cos(a_1)l_1)^2}_f + \underbrace{(y - \sin(a_1)l_1)^2}_g = l_2^2$$

$$(x+e)^2 + y^2 = l_3^2$$

$$(x+f)^2 + (y-g)^2 = l_2^2$$

$$\rightarrow x^2 + 2xe + e^2 + y^2 = l_3^2$$

$$x^2 + 2fx + f^2 + y^2 - 2yg + g^2 = l_2^2$$



$$y^2 = l_3^2 - e^2 - 2xe - x^2$$

$$\cancel{x^2} + 2fx + f^2 + l_3^2 - e^2 - 2xe - \cancel{x^2} - 2yg + g^2 = l_2^2$$

$$2fx + f^2 + l_3^2 - e^2 - 2xe - 2yg + g^2 = l_2^2$$

$$2fx + f^2 + l_3^2 - e^2 - 2xe - l_2^2 + g^2 = 2yg$$

$$\left(\frac{2fx + f^2 + l_3^2 - e^2 - 2xe - l_2^2 + g^2}{2g} \right)^2 = y^2$$

$$\left(\frac{2fx + f^2 + l_3^2 - e^2 - 2xe - l_2^2 + g^2}{2g} \right)^2 = l_3^2 - e^2 - 2xe - x^2$$

$$\frac{(x(2f - 2e) + (f^2 + l_3^2 - e^2 - l_2^2 + g^2))^2}{4g^2} = l_3^2 - e^2 - 2xe - x^2$$

$$\frac{x^2(2f-2e)^2 + 2x(2f-2e)(f^2+l_3^2-e^2-l_2^2+g^2) + (f^2+l_3^2-e^2-l_2^2+g^2)^2}{4g^2}$$

$$x^2 \left(\frac{(2f-2e)^2}{4g^2} \right) + x \left(\frac{2(2f-2e)(f^2+l_3^2-e^2-l_2^2+g^2)}{4g^2} \right) + \frac{(f^2+l_3^2-e^2-l_2^2+g^2)^2}{4g^2}$$

$$= l_3^2 - e^2 - 2xe - x^2$$

$$x^2 \left(\frac{(2f-2e)^2}{4g^2} + 1 \right) + x \left(\frac{2(2f-2e)(f^2+l_3^2-e^2-l_2^2+g^2)}{4g^2} + 2e \right) + \frac{(f^2+l_3^2-e^2-l_2^2+g^2)^2}{4g^2} + e^2 - l_3^2 = 0$$

↑
x values where circles
have an intersection

$$(x+l_4)^2 + y^2 = l_3^2$$

$$y = \pm \sqrt{l_3^2 - (x+l_4)^2}$$

↑
x on circle D

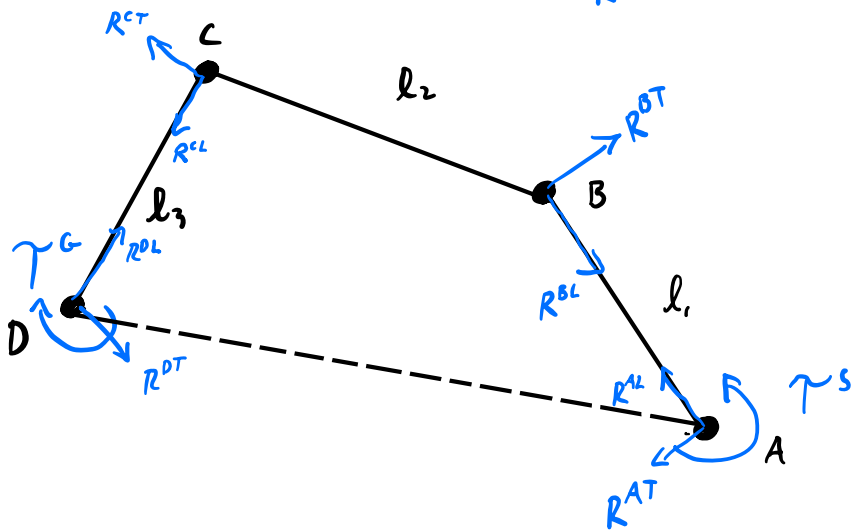
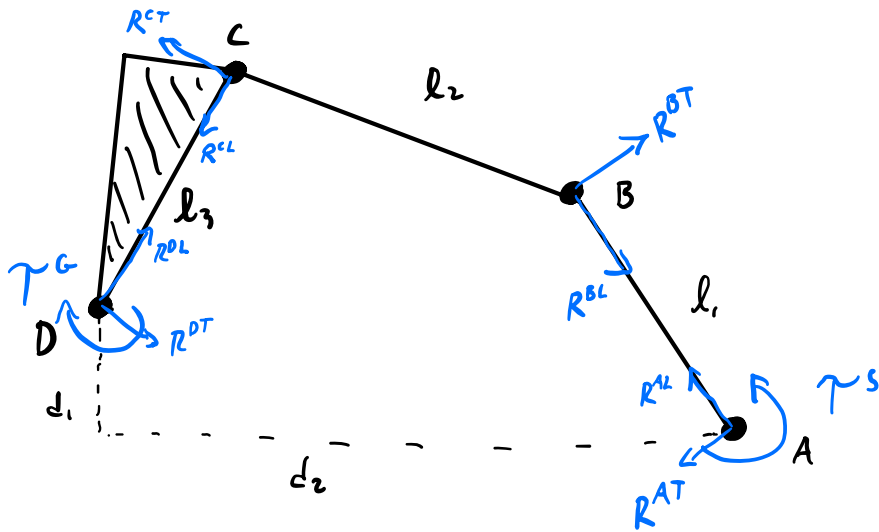
$$(x+f)^2 + (y-g)^2 = l_2^2 - (x+f)^2$$

$$y^2 - 2gy + g^2 - l_2^2 + (x+f)^2 = 0$$

$$\frac{2g \pm \sqrt{4g^2 - 4(g^2 - l_2^2 + (x+f)^2)}}{2} = y$$

↑
x on circle B

Solving For Torque-to-Torque Ratio from the Servo Joint to the Gimble Joint



$$R^{BT} l_1 = \tau^s$$

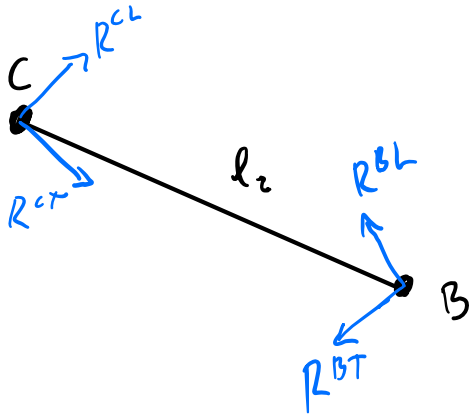
$$R^{CT} l_3 = \tau^g$$

$$R^{DT} = R^{CT}$$

$$R^{AT} = R^{BT}$$

$$R^{DL} = R^{CL}$$

$$R^{AL} = R^{BL}$$



$$\sum F_{e_1} = \cos(90^\circ - \alpha^D) R^{CT} + \cos(\alpha^D) R^{CL} - \cos(90^\circ - \alpha^A) R^{BT} - \cos(\alpha^A) R^{BL} = 0$$

$$\sum F_{e_2} = -\sin(90^\circ - \alpha^D) R^{CT} + \sin(\alpha^D) R^{CL} - \sin(90^\circ - \alpha^A) R^{BT} + \sin(\alpha^A) R^{BL} = 0$$

$$\sum M^C = -\cos(90^\circ - \alpha^A) R^{BT} \sin(\phi) l_2 - \cos(\alpha^A) R^{BL} \sin(\phi) l_2$$

R^{BL} must cancel the moment about C created by R^{BT}

$$- \sin(90^\circ - \alpha^A) R^{BT} \cos(\phi) l_2 + \sin(\alpha^A) R^{BL} \cos(\phi) l_2 = 0$$

$$\sum M^B = -\sin(\alpha^D) R^{CL} \cos(\phi) l_2 + \sin(90^\circ - \alpha^D) R^{CT} \cos(\phi) l_2$$

$$- \cos(\alpha^D) R^{CL} \sin(\phi) l_2 - \cos(90^\circ - \alpha^D) R^{CT} \sin(\phi) l_2 = 0$$

unknowns: R^{CL} , R^{CT} , R^{BL}

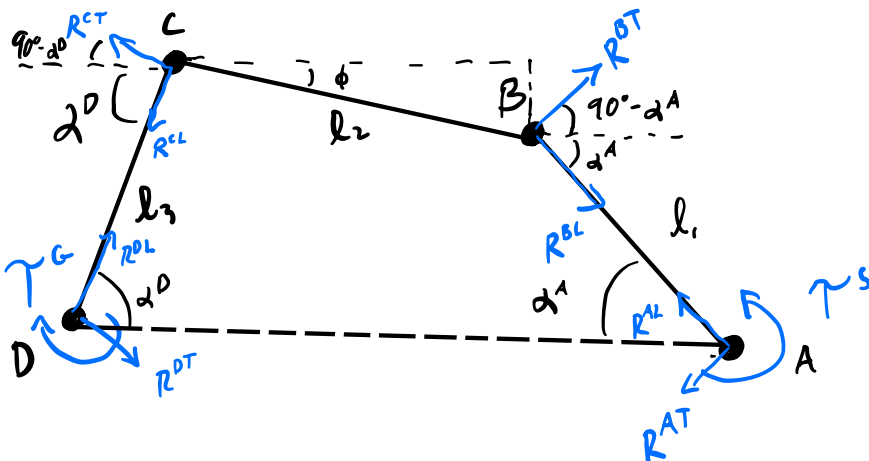
above should solve for all forces, then use

$$\tau^G = l_3 R^{CT}$$

and

$$\tau^S = l_1 R^{BT}$$

to solve for $\tau^S \rightarrow \tau^G$



$$\sum F_{e_1} = \cos(90^\circ - \alpha^p) R^{CT} + \cos(\alpha^p) R^{CL} - \cos(90^\circ - \alpha^A) R^{BT} - \cos(\alpha^A) R^{BL} = 0$$

$$\sum F_{e_2} = -\sin(90^\circ - \alpha^p) R^{CT} + \sin(\alpha^p) R^{CL} - \sin(90^\circ - \alpha^A) R^{BT} + \sin(\alpha^A) R^{BL} = 0$$

$$\begin{aligned} \sum M^C = & -\cos(90^\circ - \alpha^A) R^{BT} \sin(\phi) l_2 - \cos(\alpha^A) R^{BL} \sin(\phi) l_2 \\ & - \sin(90^\circ - \alpha^A) R^{BT} \cos(\phi) l_2 + \sin(\alpha^A) R^{BL} \cos(\phi) l_2 = 0 \end{aligned}$$

$$-\cos(90^\circ - \alpha^A) R^{BT} \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) R^{BT} \cos(\phi) l_2 =$$

$$= \cos(\alpha^A) R^{BL} \sin(\phi) l_2 - \sin(\alpha^A) R^{BL} \cos(\phi) l_2$$

$$R^{BT} \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} = R^{BL}$$

$$\begin{aligned}
& -\sin(90^\circ - \alpha^0) R^{CT} + \tan(\alpha^0) \cos(90^\circ - \alpha^A) R^{BT} + \\
& + \tan(\alpha^0) \cos(\alpha^A) \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} R^{BT} \\
& - \tan(\alpha^0) \cos(90^\circ - \alpha^0) R^{CT} - \sin(90^\circ - \alpha^A) R^{BT} + \\
& + \sin(\alpha^A) \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} R^{BT} = 0
\end{aligned}$$

$$-\sin(90^\circ - \alpha^0) R^{CT} - \tan(\alpha^0) \cos(90^\circ - \alpha^0) R^{CT} =$$

$$= -\tan(\alpha^0) \cos(90^\circ - \alpha^A) R^{BT}$$

$$- \tan(\alpha^0) \cos(\alpha^A) \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} R^{BT}$$

$$+ \sin(90^\circ - \alpha^A) R^{BT}$$

$$- \sin(\alpha^A) \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} R^{BT}$$

$$\begin{aligned}
 R^{CT} = & \left(-\tan(\alpha^D) \cos(90^\circ - \alpha^A) R^{BT} \right. \\
 & - \tan(\alpha^D) \cos(\alpha^A) \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} R^{BT} \\
 & + \sin(90^\circ - \alpha^A) R^{BT} \\
 & \left. - \sin(\alpha^A) \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} R^{BT} \right) \\
 \hline
 & - \sin(90^\circ - \alpha^D) - \tan(\alpha^D) \cos(90^\circ - \alpha^D)
 \end{aligned}$$

$$\begin{aligned}
 \gamma^G = & l_2 \left(-\tan(\alpha^D) \cos(90^\circ - \alpha^A) \frac{\gamma^S}{l_1} \right. \\
 & - \tan(\alpha^D) \cos(\alpha^A) \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} \frac{\gamma^S}{l_1} \\
 & + \sin(90^\circ - \alpha^A) \frac{\gamma^S}{l_1} \\
 & \left. - \sin(\alpha^A) \frac{-\cos(90^\circ - \alpha^A) \sin(\phi) l_2 - \sin(90^\circ - \alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} \frac{\gamma^S}{l_1} \right) \\
 \hline
 & - \sin(90^\circ - \alpha^D) - \tan(\alpha^D) \cos(90^\circ - \alpha^D)
 \end{aligned}$$

$$\begin{aligned}
 \gamma^6 = & l_3 \left(-\tan(\alpha^0) \sin(\alpha^4) \frac{\gamma^s}{l_1} \right. \\
 & - \tan(\alpha^0) \cos(\alpha^4) \frac{-\sin(\alpha^4) \sin(\phi) l_2 - \cos(\alpha^4) \cos(\phi) l_2}{\cos(\alpha^4) \sin(\phi) l_2 - \sin(\alpha^4) \cos(\phi) l_2} \frac{\gamma^s}{l_1} \\
 & + \cos(\alpha^4) \frac{\gamma^s}{l_1} \\
 & \left. - \sin(\alpha^4) \frac{-\sin(\alpha^4) \sin(\phi) l_2 - \cos(\alpha^4) \cos(\phi) l_2}{\cos(\alpha^4) \sin(\phi) l_2 - \sin(\alpha^4) \cos(\phi) l_2} \frac{\gamma^s}{l_1} \right) \\
 \hline
 = & \cos(\alpha^0) - \tan(\alpha^0) \sin(\alpha^0)
 \end{aligned}$$

$$\tau^G = \tau^S R^T$$

$\sim$ torque transformation ratio

$$R^T = \frac{l_3}{l_1} \left(-\tan(\alpha^0) \sin(\alpha^A) \right.$$

$$- \tan(\alpha^0) \cos(\alpha^A) \frac{-\sin(\alpha^A) \sin(\phi) l_2 - \cos(\alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2}$$

$$+ \cos(\alpha^A)$$

$$- \sin(\alpha^A) \frac{-\sin(\alpha^A) \sin(\phi) l_2 - \cos(\alpha^A) \cos(\phi) l_2}{\cos(\alpha^A) \sin(\phi) l_2 - \sin(\alpha^A) \cos(\phi) l_2} \Bigg)$$

$$- \cos(\alpha^0) \quad - \tan(\alpha^0) \sin(\alpha^0)$$